

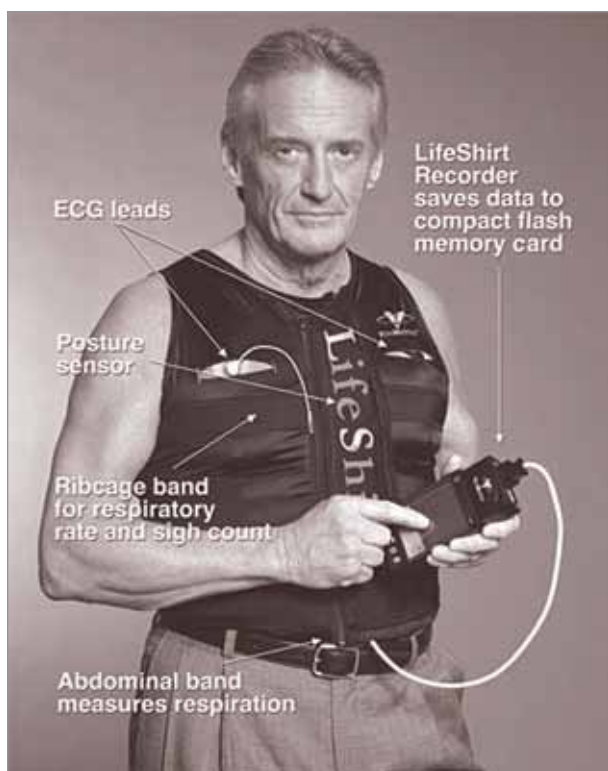


Fig. 8: The LifeShirt that monitors the wearer's vital signs
(Source: www.virtualworldlets.net)

Latitude. Pacemakers and defibrillators manufactured by Boston Scientific are available for use with their automatic latitude home monitoring system. However, they do not conduct continuous monitoring; they send the information automatically or prompt the patient to send it at the scheduled times. They can also monitor and send weight and blood pressure data to a secure website. The system delivers notifications at two levels—Red alerts and Yellow alerts. Security protocols are used to protect the medical information and only authorised people have access through a password.

Advantages of remote monitoring

The most significant benefit of remote monitoring is that it almost eliminates the routine visits to the doctor and substantially improves the survival rate of the patients. The system can detect in real time the cardiac events, abnormal heart

rhythms, and any problems with the functioning of the implanted devices—allowing the healthcare team to respond faster.

Remote monitoring helps in reducing the volume of in-office patients, allowing healthcare team to attend to more patients. Hospitalisation need is reduced, improving the quality of life and allowing improved patient management by the healthcare team.

Studies carried out over a period of five years have shown that the people using home monitoring had a

50% relative risk reduction compared to the people making regular follow-up visits to the clinics. Also, in case of patients using implanted pacemakers and a smart device for remote monitoring, there was a significantly higher rate of scheduled transmission success compared with the traditional monitoring technologies.

It was observed that some people, especially, old and aged, found it cumbersome to set up the system on their own and needed technical assistance. Also, people travelling for a long period of time may find it inconvenient to carry the monitor with them.

Conclusion

Remote monitoring of cardiovascular implantable electronic devices was developed to complement the compulsory calendar based in-person evaluations and provide convenience to the patients as well as clinics. It has the potential to

reshape monitoring of the patients as it substantially reduces the number of in-clinic visits without compromising patient safety, makes possible an early detection of clinically actionable events, and detects any malfunctioning of the implanted devices. Some systems can even connect with an ECG machine, a weighing scale, and a blood-pressure monitor.

The technology greatly assists the healthcare professionals in better managing the patients as well as the devices. Remote monitoring makes it possible to increase the number of patients that can be attended to without adding extra resources. Its use results in reduced expenses in follow-ups, possible admission, and treatment in a hospital, as well as the expenses incurred in consultation fees and transportation. There is time saving since travelling time is eliminated and there is no waiting period at the clinic for consultation.

IEDs compatible with Web based, remote monitoring systems are readily available in the market and almost all are proprietary in nature. The current technology does not allow remote reprogramming of devices, and the different proprietary systems remain a major obstacle to data integration. Ensuring the personal medical data security is another potential problem as the data is vulnerable to hacking.

These issues are likely to be resolved with future technological developments. Vivometrics LifeShirt is a non-invasive ambulatory monitoring garment with embedded sensors, which is used to monitor up to 40 physiological parameters, such as heart rate, breathing, and lung capacity. Today's devices have already far outpaced the capabilities of trans-telephonic monitoring and continuous development is taking place to enable them improve healthcare and at a lower cost. **EFY**